

Sheng-An (Samuel) Xu

Ph.D. Student in Industrial Engineering and Operations Research | University of California, Berkeley

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Research: Large-scale optimization, online learning, and generative models.

Education

University of California, Berkeley

Aug. 2025–present

Ph.D. student, Industrial Engineering and Operations Research; advisor: Prof. Ying Cui
M.S., Industrial Engineering and Operations Research (May 2026).

Nanjing University

Sep. 2021–Jul. 2025

B.S., Mathematics and Applied Mathematics

GPA: 4.66/5.00; rank: 1/48 in major, 2/132 in department; WES evaluation: 3.94/4.00.

University of Wisconsin–Madison

Jan.–May 2024

Exchange student; GPA: 4.0/4.0

Publications

Sheng-An Xu, Hanyang Li, Jianhao Ma, and Ying Cui. *Beyond Shadow Weights: Quantization-Aware Training as Quantized-Endpoint Descent.*

Preprint, 2026. arXiv:2609.21834.

Yu-Jie Zhang, Sheng-An Xu, Peng Zhao, and Masashi Sugiyama. *Generalized Linear Bandits: Almost Optimal Regret with One-Pass Update.*

NeurIPS 2025. arXiv:2507.11847.

Research Experience

Statistical and Computational Efficiency of Nonlinear Bandits

Jan.–May 2025

LAMDA Group, Nanjing University; advisor: Prof. Peng Zhao

- Designed generalized linear bandit algorithms with one-pass online updates, replacing maximum likelihood estimation with online mirror descent for computationally efficient confidence regions.
- Used mix-loss analysis and self-concordance to sharpen regret bounds and reduce dependence on a condition number.

Optimization Based on Diffusion Models

Apr.–Nov. 2024

Nanjing University; advisor: Prof. Caihua Chen

- Trained conditional diffusion models on optimization trajectories to generate feasible solutions for parameterized integer programs.
- Explored discrete diffusion and heuristic sampling for graph-structured optimization problems.

Professional Experience

Machine Learning Engineer Intern

Jun.–Aug. 2024

Nanjing Huihu Intelligence Technology Co., Ltd.

- Built a streaming speaker-separation and subtitle pipeline; formulated audio recovery as a composite optimization problem and applied accelerated proximal gradient methods.
- Applied optical flow, SlowFast, and YOLOv10 to video analysis, achieving over 80% scenario-classification accuracy.

Selected Honors and Awards

- National Scholarship, top 0.2% nationally (2021–22).
- Guo-Xie-Birong Scholarship, department’s highest honor (2023–24).
- Jiangsu Province Government Scholarship for Overseas Study (2025).
- Special Scholarship of Basic Sciences (2022, 2023, 2024).
- Meritorious Prize, Mathematical Contest in Modeling (2023).
- Outstanding Leader, Tian-Jian Voluntary Association (2022–23).

Skills

Programming: C++, Python, MATLAB, R. **Tools:** Git, LaTeX, Mathematica, MySQL.